

Lab 3 - Routing Protocol

NAME: LAU YEE WEN

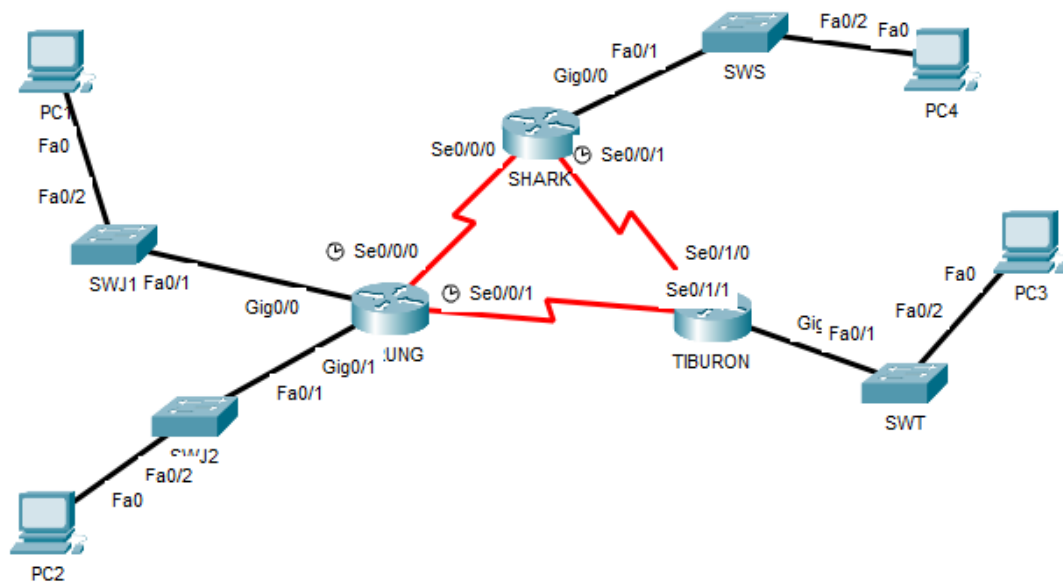
MATRIC NUM.: A23CS0099

SECTION: 12

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.1	255.255.255.224	-
4		G0/1	172.18.110.33	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.129	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.65	255.255.255.192	-
11	PC1	-	172.18.110.30	255.255.255.224	172.18.110.1
12	PC2	-	172.18.110.62	255.255.255.224	172.18.110.33
13	PC3	-	172.18.110.190	255.255.255.192	172.18.110.129
14	PC4	-	172.18.110.126	255.255.255.192	172.18.110.65

LAB TASKS

Task 1 – IP Addressing

- Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.

- JERUNG LAN1 (20 hosts): /27 (32 IPs, 30 usable IPs) → 172.18.110.0/27 (**172.18.110.1-172.18.110.30**)
 $2^5 = 32$
 Subnet mask: /27
 Usable IPs: $32 - 2 = 30$
- JERUNG LAN2 (30 hosts): /27 (32 IPs, 30 usable IPs) → 172.18.110.32/27 (**172.18.110.33-172.18.110.62**)
 $2^5 = 32$
 Subnet mask: /27
 Usable IPs: $32 - 2 = 30$
- SHARK LAN (60 hosts): /26 (64 IPs, 62 usable IPs) → 172.18.110.64/26 (**172.18.110.65-172.18.110.126**)
 $2^6 = 64$
 Subnet mask: /26
 Usable IPs: $64 - 2 = 62$
- TIBURON LAN (50 hosts): /26 (64 IPs, 62 usable IPs) → 172.18.110.128/26 (**172.18.110.129- 172.18.110.190**)
 $2^6 = 64$
 Subnet mask: /26
 Usable IPs: $64 - 2 = 62$
- Point-to-Point Links** (JERUNG-SHARK, JERUNG-TIBURON, SHARK-TIBURON): 2 hosts each → /30 (4 IPs, 2 usable IPs).
 - JERUNG-SHARK: 172.18.110.192/30 → **Usable: (172.18.110.193 – 172.18.110.194)**
 - JERUNG-TIBURON: 172.18.110.196/30 → **Usable: 172.18.110.197 – 172.18.110.198**
 - SHARK-TIBURON: 172.18.110.200/30 → **Usable: 172.18.110.201 – 172.18.110.202** $2^2 = 4$
 Subnet mask: /30
 Usable IPs: $4 - 2 = 2$

Device	Subnetwork	Prefix	Subnet Mask	Network Address	Usable IP Range
JERUNG	LAN1	/27	11111111. 11111111. 11111111. 11100000 =255.255.255.224	172.18.110.0	172.18.110.1 - 172.18.110.30
	LAN2	/27	11111111. 11111111. 11111111. 11100000 =255.255.255.224	172.18.110.32	172.18.110.33 - 172.18.110.62
SHARK	LAN	/26	11111111. 11111111. 11111111. 11000000 =255.255.255.192	172.18.110.64	172.18.110.65 - 172.18.110.126
TIBURON	LAN	/26	11111111. 11111111. 11111111. 11000000 =255.255.255.192	172.18.110.128	172.18.110.129 - 172.18.110.190
Connections	JERUNG-SHARK	/30	11111111. 11111111. 11111111. 11111100 =255.255.255.252	172.18.110.192	172.18.110.193 - 172.18.110.194
	JERUNG-TIBURON	/30	11111111. 11111111. 11111111. 11111100 =255.255.255.252	172.18.110.196	172.18.110.197 - 172.18.110.198
	SHARK-TIBURON	/30	11111111. 11111111. 11111111. 11111100 =255.255.255.252	172.18.110.200	172.18.110.201 - 172.18.110.202

2. Using the IP addresses calculated, configure the devices with the appropriate information.

JERUNG:

The screenshot shows the JERUNG network configuration window. The 'Config' tab is selected, and the 'INTERFACE' section is expanded. The configuration for GigabitEthernet0/0 is displayed:

- Port Status:** On (checked)
- Bandwidth:** 100 Mbps (selected)
- Duplex:** Full Duplex (selected)
- MAC Address:** 0002.1636.2A01
- IP Configuration:**
 - IPv4 Address: 172.18.110.1
 - Subnet Mask: 255.255.255.224
- Tx Ring Limit:** 10

JERUNG

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

GigabitEthernet0/1

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0002.1636.2A02

IP Configuration

IPv4 Address 172.18.110.33

Subnet Mask 255.255.255.224

Tx Ring Limit 10

JERUNG

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/0/0

Port Status

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 172.18.110.193

Subnet Mask 255.255.255.252

Tx Ring Limit 10

JERUNG

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/0/1

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 172.18.110.197

Subnet Mask 255.255.255.252

Tx Ring Limit 10

TIBURON:

TIBURON

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0**
- GigabitEthernet0/1
- Serial0/1/0
- Serial0/1/1

GigabitEthernet0/0

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00D0.BC84.3401

IP Configuration

IPv4 Address 172.18.110.129

Subnet Mask 255.255.255.192

Tx Ring Limit 10

TIBURON

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0
- GigabitEthernet0/1
- Serial0/1/0**
- Serial0/1/1

Serial0/1/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 172.18.110.202

Subnet Mask 255.255.255.252

Tx Ring Limit 10

TIBURON

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/1/0

Serial0/1/1

Serial0/1/1

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 172.18.110.198

Subnet Mask 255.255.255.252

Tx Ring Limit 10

SHARK:

SHARK

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

GigabitEthernet0/0

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0040.0B2C.0A01

IP Configuration

IPv4 Address 172.18.110.65

Subnet Mask 255.255.255.192

Tx Ring Limit 10

SHARK

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0
- GigabitEthernet0/1
- Serial0/0/0**
- Serial0/0/1

Serial0/0/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 172.18.110.194

Subnet Mask 255.255.255.252

Tx Ring Limit 10

SHARK

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0
- GigabitEthernet0/1
- Serial0/0/0
- Serial0/0/1**

Serial0/0/1

Port Status ☒ On

Duplex ☐ Full Duplex

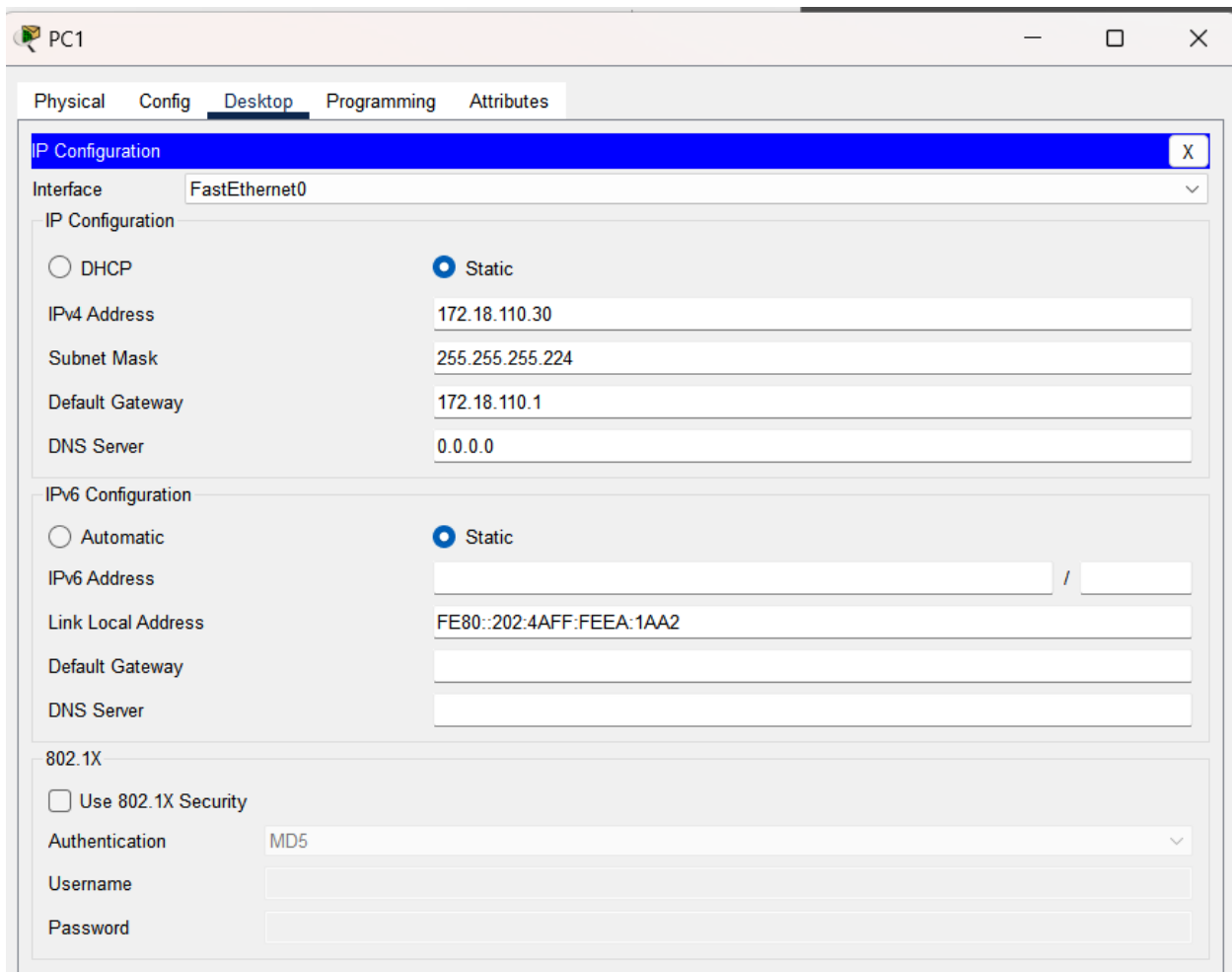
Clock Rate 2000000

IP Configuration

IPv4 Address 172.18.110.201

Subnet Mask 255.255.255.252

Tx Ring Limit 10

PC1:

The screenshot shows a window titled "PC1" with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there are five tabs: "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is currently selected and highlighted. Below the tabs, there is a section titled "IP Configuration" with a blue header bar and a close button (X) on the right. Under this section, the "Interface" is set to "FastEthernet0". The "IP Configuration" sub-section has two radio buttons: "DHCP" (unselected) and "Static" (selected). Below these are four text input fields: "IPv4 Address" (172.18.110.30), "Subnet Mask" (255.255.255.224), "Default Gateway" (172.18.110.1), and "DNS Server" (0.0.0.0). The "IPv6 Configuration" sub-section also has two radio buttons: "Automatic" (unselected) and "Static" (selected). Below these are four text input fields: "IPv6 Address" (empty), "Link Local Address" (FE80::202:4AFF:FEEA:1AA2), "Default Gateway" (empty), and "DNS Server" (empty). The "802.1X" sub-section has a checkbox "Use 802.1X Security" (unchecked). Below it are three text input fields: "Authentication" (MD5), "Username" (empty), and "Password" (empty).

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.18.110.30

Subnet Mask 255.255.255.224

Default Gateway 172.18.110.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::202:4AFF:FEEA:1AA2

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC2:

The screenshot shows a configuration window for PC2 with the following tabs: Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying the IP Configuration settings for the FastEthernet0 interface.

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 172.18.110.62

Subnet Mask: 255.255.255.224

Default Gateway: 172.18.110.33

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::2E0:8FFF:FED8:CCD1

Default Gateway:

DNS Server:

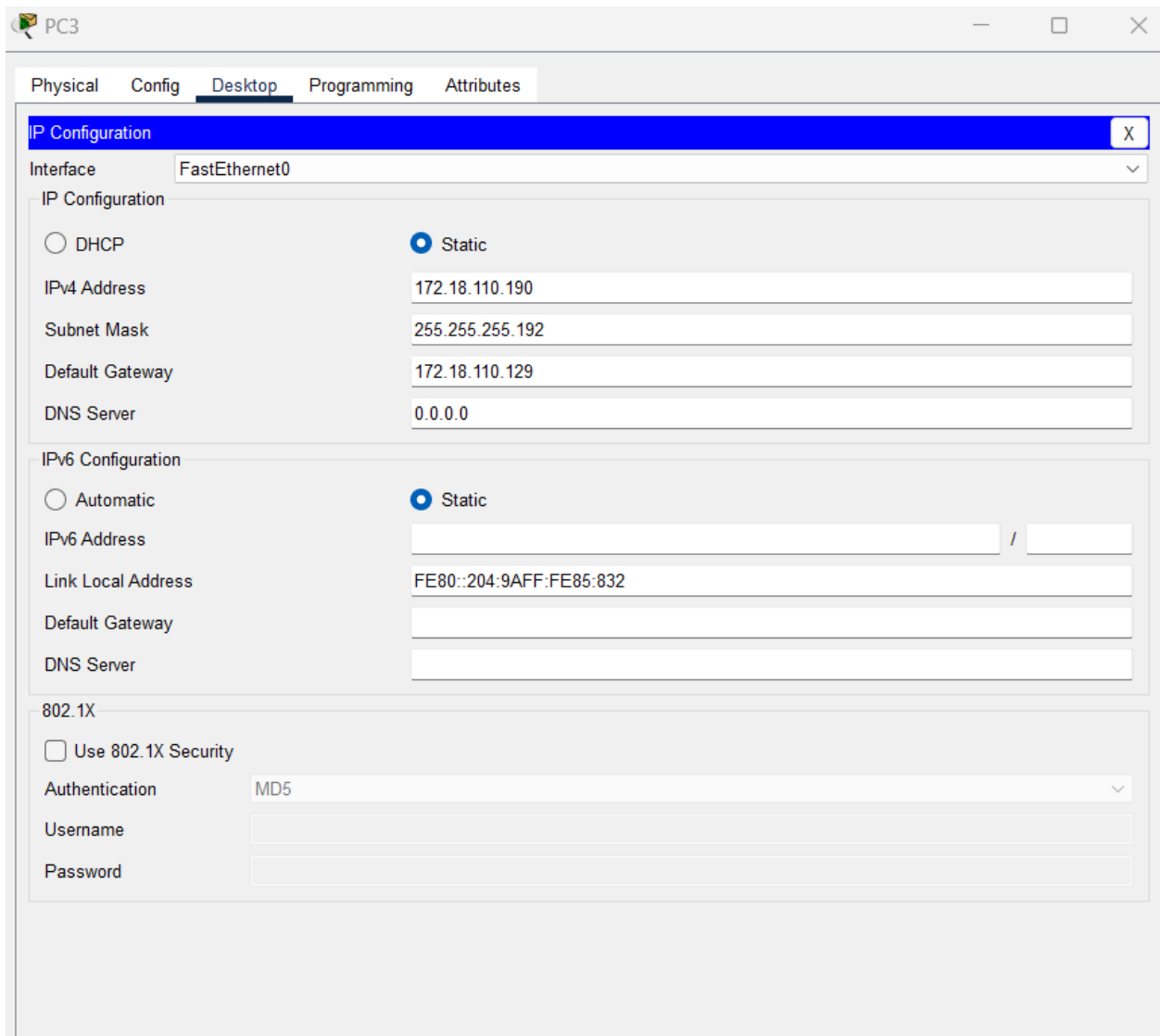
802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

PC3:

The screenshot shows a configuration window for PC3 with the following tabs: Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, and the IP Configuration window is open. The interface is set to FastEthernet0. Under IP Configuration, the Static option is selected. The IPv4 Address is 172.18.110.190, Subnet Mask is 255.255.255.192, Default Gateway is 172.18.110.129, and DNS Server is 0.0.0.0. Under IPv6 Configuration, the Static option is also selected. The IPv6 Address field is empty, the Link Local Address is FE80::204:9AFF:FE85:832, and the Default Gateway and DNS Server fields are empty. Under 802.1X, the Use 802.1X Security checkbox is unchecked, and the Authentication is set to MD5. The Username and Password fields are empty.

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.18.110.190

Subnet Mask 255.255.255.192

Default Gateway 172.18.110.129

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::204:9AFF:FE85:832

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC4:

PC4

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.18.110.126

Subnet Mask 255.255.255.192

Default Gateway 172.18.110.65

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::200:CFF:FEAA:16C8

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.
 - a. CLI
 - i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

- b. PT tool
- Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

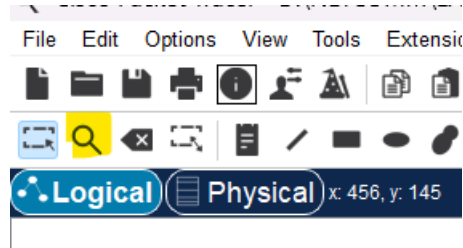


Figure B

A screenshot of the 'Routing Table for SHARK' window. The table has five columns: Type, Network, Port, Next Hop IP, and Metric. It contains two entries: a Connected (C) entry for 172.18.110.0/26 and a Local (L) entry for 172.18.110.1/32, both with GigabitEthernet0/0 as the port and 0/0 as the metric.

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

JERUNG**CLI:**
 JERUNG

Physical Config **CLI** Attributes

IOS Command Line Interface

```

JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/0
JERUNG(config-if)#ip address 172.18.110.193 255.255.255.252
JERUNG(config-if)#ip address 172.18.110.193 255.255.255.252
JERUNG(config-if)#no shutdown
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/0
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/1
JERUNG(config-if)#ip address 172.18.110.197 255.255.255.252
JERUNG(config-if)#ip address 172.18.110.197 255.255.255.252
JERUNG(config-if)#no shutdown
JERUNG(config-if)#exit
JERUNG(config)#exit
JERUNG#
%SYS-5-CONFIG_I: Configured from console by console

JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.0/27 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.32/27 is directly connected, GigabitEthernet0/1
L       172.18.110.33/32 is directly connected, GigabitEthernet0/1
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1

JERUNG#
  
```

PT Tool:

Routing Table for JERUNG				
Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

TIBURON**CLI:**
 TIBURON

Physical Config **CLI** Attributes

IOS Command Line Interface

```

TIBURON(config-if)#ip address 172.18.110.202 255.255.255.252
TIBURON(config-if)#no shutdown

TIBURON(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

TIBURON(config-if)#exit
TIBURON(config)#interface Serial0/1/0
TIBURON(config-if)#
TIBURON(config-if)#exit
TIBURON(config)#interface Serial0/1/1
TIBURON(config-if)#ip address 172.18.110.198 255.255.255.252
TIBURON(config-if)#ip address 172.18.110.198 255.255.255.252
TIBURON(config-if)#no shutd
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

TIBURON(config-if)#no shutdown
TIBURON(config-if)#exit
TIBURON(config)#exit
TIBURON#
%SYS-5-CONFIG_I: Configured from console by console

TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.128/26 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0

TIBURON#
  
```

PT Tool:

Routing Table for TIBURON					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0	
C	172.18.110.196/30	Serial0/1/1	---	0/0	
L	172.18.110.198/32	Serial0/1/1	---	0/0	
C	172.18.110.200/30	Serial0/1/0	---	0/0	
L	172.18.110.202/32	Serial0/1/0	---	0/0	

SHARK**CLI:**
 **SHARK**

Physical Config **CLI** Attributes

IOS Command Line Interface

```

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

SHARK(config-if)#
SHARK(config-if)#exit
SHARK(config)#interface Serial0/0/0
SHARK(config-if)#
SHARK(config-if)#exit
SHARK(config)#interface Serial0/0/1
SHARK(config-if)#ip address 172.18.110.201 255.255.255.252
SHARK(config-if)#ip address 172.18.110.201 255.255.255.252
SHARK(config-if)#no s
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
no shutdown
      ^
% Invalid input detected at '^' marker.

SHARK(config-if)#no shutdown
SHARK(config-if)#exit
SHARK(config)#exit
SHARK#
%SYS-5-CONFIG_I: Configured from console by console

SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1

SHARK#

```

PT Tool:

Routing Table for SHARK				
Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

2. Try to ping PC2 from PC1, paste the results here.

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Request timed out.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

4. Explain the reason(s) behind the results.

PC1 ping PC2 successfully because they are in the same subnet. However, PC1 failed to ping PC4 because they are in different subnets causing them unable to communicate directly to each other and there is no configurations that enables devices in different subnets to communicate across different router. Also, PC1 and PC4 are connected to different routers thus the result shows “Destination host unreachable”.

5. What needs to be done to ensure all PCs can ping each other successfully?

- All interfaces must have the proper IP addresses and subnet masks.
- Default gateways must be configured on all PCs.
- Verify that no access control lists (ACLs) are blocking communication.
- RIP or static routes must be configured so that routing information can be shared between routers.

Task 2 – Routing Configuration

- Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

Routing Table for TIBURON					Routing Table for JERUNG				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0	C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0	L	172.18.110.161/32	GigabitEthernet0/0	---	0/0

Figure D

Routing Table for TIBURON					Routing Table for JERUNG				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0	C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0	L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0	C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0	L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0	C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0	L	172.18.110.193/32	Serial0/0/0	---	0/0
					C	172.18.110.196/30	Serial0/0/1	---	0/0
					L	172.18.110.197/32	Serial0/0/1	---	0/0

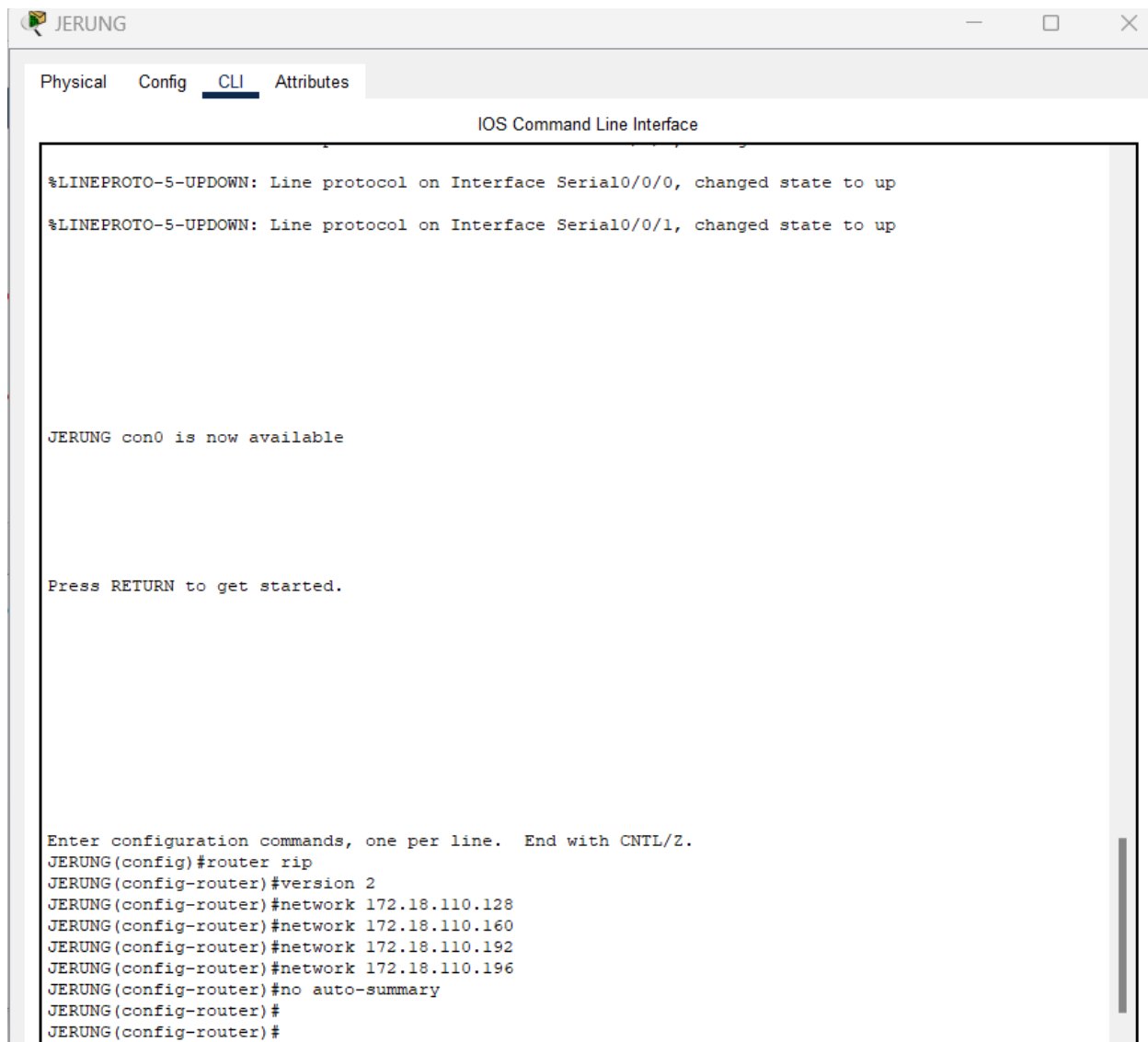
- In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

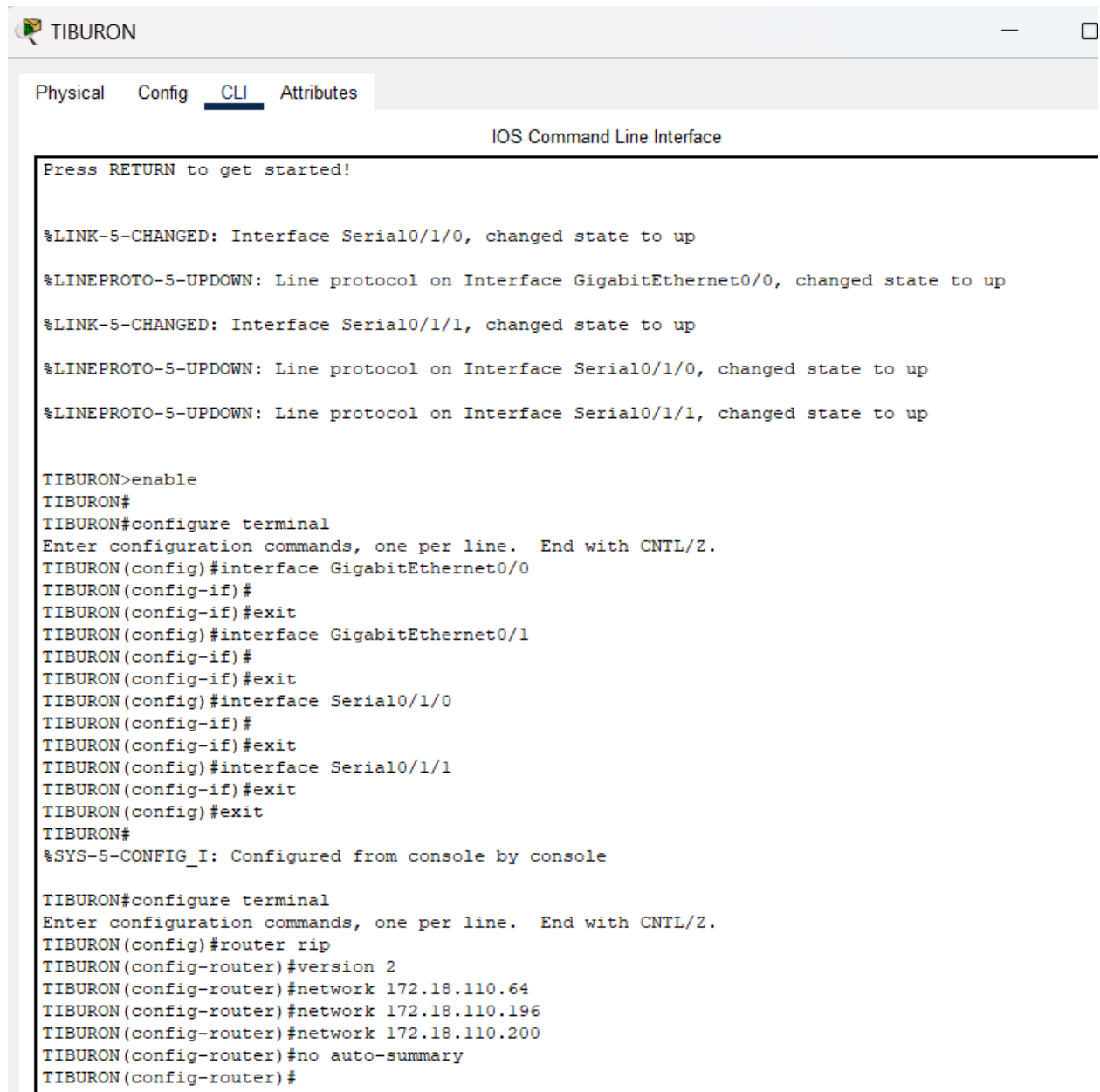
JERUNG:

- a. What can you say about the addresses used in the 'network' instructions in Figure E?
The addresses in the 'network' instructions are the starting points for different IP ranges (subnets). Adding these network addresses in RIP lets routers share information about how to reach each subnet. This allows devices in different subnets and routers to communicate with each other efficiently.
3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```

TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200

```

TIBURON:


The screenshot shows a window titled "TIBURON" with a tabbed interface. The "CLI" tab is selected, displaying the "IOS Command Line Interface". The interface shows a series of status messages and configuration commands entered in a terminal-like environment.

```

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

TIBURON>enable
TIBURON#
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#interface GigabitEthernet0/0
TIBURON(config-if)#
TIBURON(config-if)#exit
TIBURON(config)#interface GigabitEthernet0/1
TIBURON(config-if)#
TIBURON(config-if)#exit
TIBURON(config)#interface Serial0/1/0
TIBURON(config-if)#
TIBURON(config-if)#exit
TIBURON(config)#interface Serial0/1/1
TIBURON(config-if)#exit
TIBURON(config)#exit
TIBURON#
%SYS-5-CONFIG_I: Configured from console by console

TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 172.18.110.64
TIBURON(config-router)#network 172.18.110.196
TIBURON(config-router)#network 172.18.110.200
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#
  
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Routing Table for TIBURON					Routing Table for JERUNG				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1	C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1	L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0	C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0	L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1	R	172.18.110.128/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0	C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0	L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0	C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0	L	172.18.110.197/32	Serial0/0/1	---	0/0
					R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

- a. What are the changes seen in TIBURON?

Before RIP, the routing table only showed directly connected networks. After RIP, TIBURON's table includes RIP-learned routes (R) for the networks connected to JERUNG and SHARK.

- b. What are the Networks with type R in TIBURON and JERUNG?

TIBURON: 172.18.110.0/27, 172.18.110.32/27, and 172.18.110.192/30

JERUNG: 172.18.110.128/26, and 172.18.110.200/30

- c. Ping PC3 from PC1. Was it successful?

Yes

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.190: bytes=32 time=19ms TTL=126
Reply from 172.18.110.190: bytes=32 time=17ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 19ms, Average = 12ms

C:\>
```

- d. Ping PC4 from PC1. Was it successful?

Not successful

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

- e. Explain the reasons for your answer.

- **PC1 Successfully Pings PC3:** Both PCs (PC1 and PC3) are in subnets directly connected to the JERUNG router. JERUNG can route between its interfaces because it inherently knows the routes for its directly connected subnets.
- **PC1 Fails to Ping PC4:** PC1 and PC4 are in different subnets and are not directly connected to the same router. PC1 is connected to JERUNG, and PC4 is connected to SHARK. SHARK does not have RIP configured, so it cannot share or learn routing information for PC4's subnet. This lack of configuration prevents JERUNG from knowing how to route packets to PC4's network.

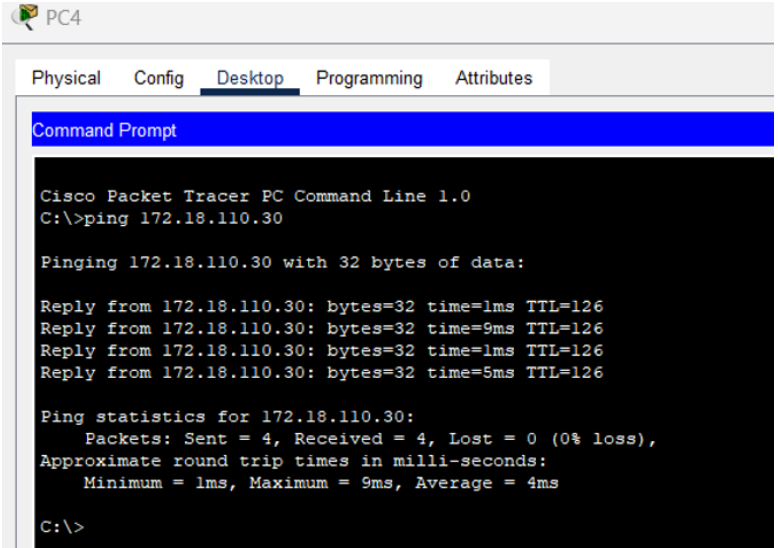
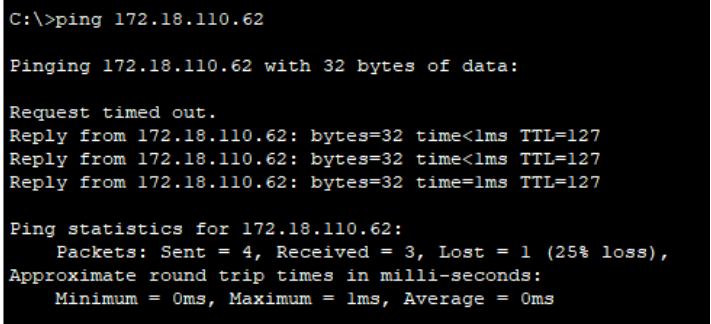
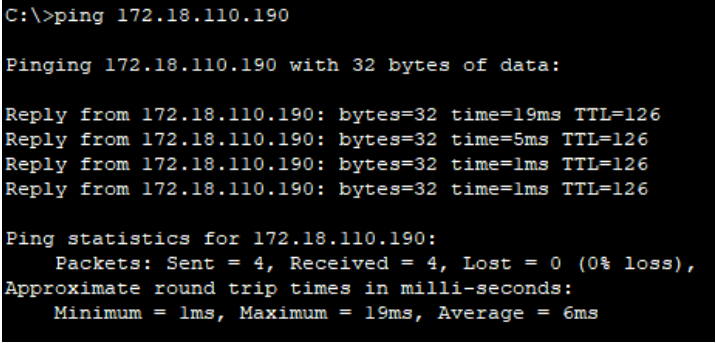
- f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.128
SHARK(config-router)#network 172.18.110.160
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.196
SHARK(config-router)#no auto-summary
SHARK(config-router)#
```

- g. Open router SHARK's routing table and paste here.

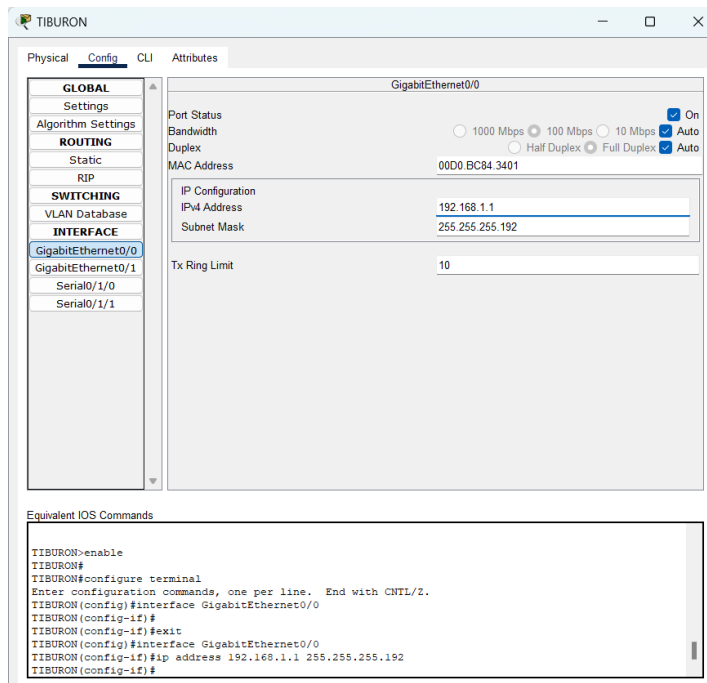
Routing Table for SHARK					
Type	Network	Port	Next Hop IP	Metric	
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1	
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1	
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/1	
C	172.18.110.192/30	Serial0/0/0	---	0/0	
L	172.18.110.194/32	Serial0/0/0	---	0/0	
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1	
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1	
C	172.18.110.200/30	Serial0/0/1	---	0/0	
L	172.18.110.201/32	Serial0/0/1	---	0/0	

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	Success  <pre> Cisco Packet Tracer PC Command Line 1.0 C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Reply from 172.18.110.30: bytes=32 time=9ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Reply from 172.18.110.30: bytes=32 time=5ms TTL=126 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 9ms, Average = 4ms C:\> </pre>
2	Success  <pre> C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Request timed out. Reply from 172.18.110.62: bytes=32 time<1ms TTL=127 Reply from 172.18.110.62: bytes=32 time<1ms TTL=127 Reply from 172.18.110.62: bytes=32 time=1ms TTL=127 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 1ms, Average = 0ms </pre>
3	Success  <pre> C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=19ms TTL=126 Reply from 172.18.110.190: bytes=32 time=5ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 19ms, Average = 6ms </pre>

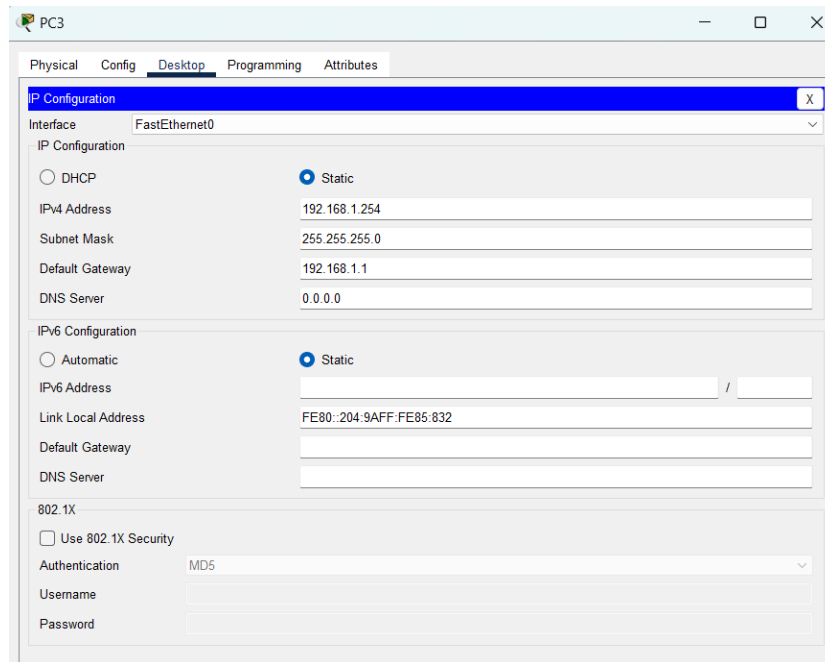
Task 3 – Routing Update

- Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.



- This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: 192.168.1.0/24
- As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.254
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1



- c. After this change, can PC4 and PC1 ping PC3?

Both PC4 and PC1 cannot ping PC3

PC1:

```
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Request timed out.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

PC4:

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 172.18.110.65: Destination host unreachable.
Request timed out.
Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>

```

- d. Copy and paste the routing tables for both SHARK and TIBURON here.

Routing Table for SHARK					Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1	R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1	R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0	R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
L	172.18.110.194/32	Serial0/0/0	---	0/0	C	172.18.110.196/30	Serial0/1/1	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1	L	172.18.110.198/32	Serial0/1/1	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1	C	172.18.110.200/30	Serial0/1/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0	L	172.18.110.202/32	Serial0/1/0	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0	C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
					L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

- e. Referring to the routing table, explain your findings.

SHARK's routing table does not include 192.168.1.0/24 subnet, whereas TIBURON's routing table includes the 192.168.1.0/24 subnet. This causes PC4 and PC1 cannot ping PC3 successfully. JERUNG and SHARK need to add the 192.168.1.0/24 subnet in their routing table so that all devices can communicate successfully across different subnets.

- f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?

Ensure all routers have the correct network statements in RIP to propagate 192.168.1.0/24. This can be done by updating RIP on TIBURON and propagate the new route for other subnets.

- g. Show your configurations in TIBURON to ensure end-to-end connectivity.

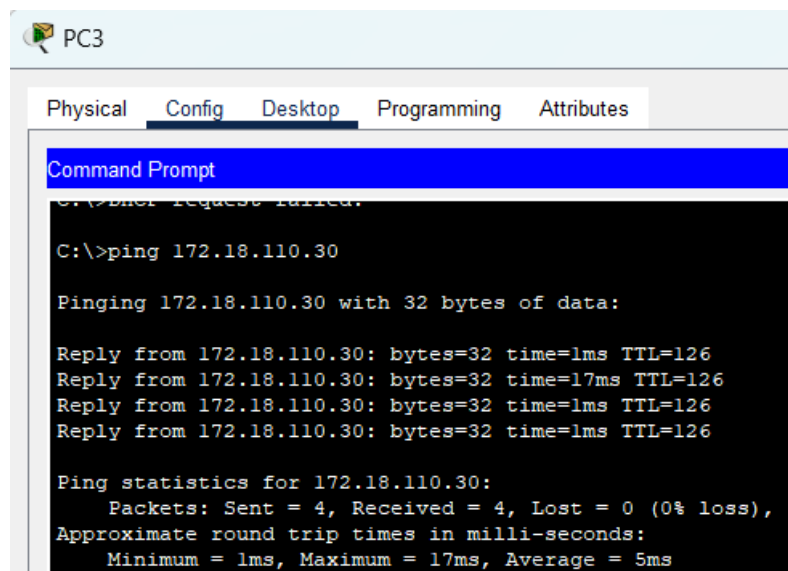
```
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#
```

- The network 192.168.1.0 command tells the router to advertise and learn routes in this subnet range.

- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result
1	Success
2	Success
4	Success

PC1



PC2

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=19ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=24ms TTL=126
Reply from 172.18.110.62: bytes=32 time=9ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 24ms, Average = 13ms
```

PC4

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.126: bytes=32 time=25ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 25ms, Average = 7ms

C:\>|
```

REFLECTION

What have you learned in this task?

From this task, I learned how to configure routers using RIP to enable communication between different subnets. I also understood the importance of subnetting for efficient IP address allocation and how dynamic protocols like RIP simplify routing between routers. Additionally, I learned how to troubleshoot connectivity issues using commands like "ping" and "show ip route." This task helped improve my understanding of routing protocols and their configuration.